

How My Thesis Works: A Self-Study Guide

From Classical Metrics to Functional Forecasting

Zeynep Bozoklu — Personal learning notes

These notes explain every concept in my thesis starting from what I already know from first-year econometrics. The goal is to be able to explain any part of the work clearly and confidently in a seminar or presentation. The actual numbers from the results appear throughout so the explanations stay grounded.

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1 Why This Problem Matters

Turkey’s electricity grid operator needs to forecast how much electricity will be consumed in each of the 24 hours of the following day. This is called a **day-ahead load forecast**. Grid operators use these forecasts to schedule power plants, manage reserves, and price electricity in the wholesale market.

The Turkish operator publishes an official forecast. My research question is simple: *can a statistical model do better?*

The answer, for the model I build: yes, by about **19% in mean absolute error** and **19% in mean absolute percentage error** relative to the official forecast.

Why might a statistical model beat the official forecast? The official forecast is itself the output of a complex proprietary system. But a statistical model trained on historical patterns can sometimes learn systematic biases or regularities that the official model misses.

2 The Economic Setting

2.1 Why the load curve has the shape it has

The 24-hour electricity load curve is a direct imprint of human behavior and economic activity. Understanding its shape is essential for explaining why the forecasting problem is hard and why the functional data approach is natural.

Overnight trough (hours 0–5). Most people are asleep. Factories running night shifts are the main consumers, along with street lighting and always-on infrastructure (servers, refrigeration, pumping stations). Demand is low and stable, varying mainly by season (heating in winter, no AC). This is the easiest part of the curve to forecast — weekly seasonality (lag 7) captures most of the variation. Our model gains 40–60% in MAPE over the official forecast at these hours.

Morning ramp-up (hours 6–8). The single most economically consequential transition in the curve. Within two hours, electricity demand roughly doubles. The drivers are: residential consumers waking up (lighting, coffee machines, showers), commercial buildings switching on HVAC, and industrial facilities starting morning shifts. The *timing* of this ramp depends critically on temperature: on a hot summer day, air conditioning turns on immediately at sunrise; on a mild day, the ramp is gentler. This is the one part of the daily curve where our model loses to the official forecast (by up to 24% MAPE at hour 8), almost certainly because the official forecast incorporates weather forecasts that we do not have access to.

Midday plateau (hours 9–16). Industrial and commercial demand is at full intensity. This part of the curve is dominated by large consumers — manufacturing, mining, construction — whose behavior is predictable from the day of week and seasonal calendar. Our model wins comfortably here (20–35% MAPE gain), suggesting these systematic patterns are well captured by holiday indicators, weekend flags, and lagged load-shape features.

Evening surge (hours 17–19). The second major transition. Industrial activity begins to wind down while residential demand surges: people come home, turn on televisions, cook dinner, and in summer run air conditioning at maximum. In Turkey this peak coincides with the hottest part of the evening in summer and with heating demand in winter. The official forecast is worst here: it underpredicts by 614–658 MWh on average at hours 17–19, substantially larger than at any other time of day. Our model still loses at hours 18–19 but with a much smaller bias.

Night wind-down (hours 20–23). Residential demand falls as people go to sleep. Industrial activity has mostly stopped. The curve declines smoothly toward the overnight trough. Our

model wins by 17–20% MAPE here.

Why this matters for FDA: the daily load curve is not 24 independent numbers — it is a single coherent trajectory shaped by overlapping behavioral rhythms. The morning ramp, midday plateau, and evening peak are features of the *whole curve*, not of individual hours. Modeling the curve as a functional object rather than a vector of 24 hourly values lets us directly work with these shape features (via FPCA components and derivative summaries) instead of treating each hour in isolation.

2.2 Turkey’s electricity market: who uses these forecasts

Turkey liberalised its electricity market through a series of reforms from 2001 to 2013. The key institutions are:

TEİAŞ (Türkiye Elektrik İletim A.Ş.) — the transmission system operator (TSO). It owns and operates the high-voltage grid and is responsible for system balance: at every moment, generation must equal consumption. TEİAŞ publishes the official day-ahead load forecast, which is the benchmark in this thesis.

EPIAS (Enerji Piyasaları İşletme A.Ş.) — the energy exchange that operates the day-ahead market (DAM) and intraday market. The DAM clears at noon the day before delivery, with generators and retailers submitting hourly bids and offers.

Generators submit supply bids for each hour of the following day. The clearing price depends on the demand they expect: if demand is forecast to be high in the evening, peaking plants will be called on, driving up prices. Generators with better demand forecasts can time their bids more strategically — bidding higher when they know demand will be tight, and backing off when it will be slack.

Retailers and large industrial consumers submit demand bids and purchase power at DAM prices. A large factory with a better load forecast of its own consumption faces smaller imbalance penalties in the real-time balancing market. Inaccurate internal forecasts expose them to buying expensive balancing power.

The TSO’s role in scheduling: TEİAŞ uses its load forecast to run *unit commitment* — deciding which power plants to start up and keep running. Starting a thermal plant (coal, gas) takes hours and has significant startup costs. A 19% better forecast means TEİAŞ can commit capacity more efficiently: fewer unnecessary startups and fewer costly real-time reserve activations.

The economic value of better forecasting. A 1% improvement in MAPE is estimated to reduce balancing costs by millions of euros per year in medium-sized European electricity markets (Bunn & Karakatsani, 2003). Turkey’s electricity consumption is around 350 TWh/year and growing. Our 18.7% MAPE improvement, if used operationally, could represent substantial savings in reserve procurement and balancing costs — the exact amount would require knowledge of Turkey’s balancing market prices, but the order of magnitude is economically meaningful.

2.3 The systematic underprediction puzzle

The official forecast underpredicts actual consumption at every single hour of the day, every year in the evaluation period. The numbers:

Year	Official bias (MWh)	Main model bias (MWh)
2023	+344	+176
2024	+585	+34
2025	+472	+98
Average	+467	+103

Positive bias means actual > forecast, i.e., the forecast is systematically too low.

Three observations stand out:

1. The underprediction worsens in 2024 (+585 MWh) before partially recovering in 2025. This coincides with an exceptionally hot summer in Turkey in 2024 driving unusually high air conditioning demand — a shock the official model apparently did not absorb.
2. The underprediction is largest at evening hours (17–20), with a bias of 614–658 MWh. This is the residential demand surge that is hardest to predict from institutional/commercial patterns alone.
3. Our model’s bias is much smaller (34 MWh in 2024, essentially zero) because the rolling-origin design continuously re-trains on the most recent data. As Turkish demand grew in 2024, our model adapted automatically.

2.4 Economic incentives and agent behavior

Why would the official forecaster persistently underpredict? Several mechanisms from the economics of forecasting apply here.

Asymmetric loss functions. If the TSO faces asymmetric costs — over-procurement of capacity (scheduling too many plants and leaving them idle) is more politically or contractually costly than under-procurement (buying expensive real-time reserves) — then rational forecast optimisation under asymmetric loss produces a downward-biased forecast. This is a well-known result: when the loss from over-forecasting exceeds the loss from under-forecasting, the optimal point forecast is below the conditional mean.

Incentive to understate demand growth. A TSO that continuously reports “demand exceeded forecast” may face less scrutiny than one that “wasted money on excess capacity.” Politically, underpredicting and then being surprised by demand growth can be framed as evidence of a strong economy. Overpredicting and having idle power plants on contract is visible and costly in a different way.

Model staleness. Official forecasting models at large institutions are updated infrequently — they require regulatory approval, internal audits, and systems integration. A model calibrated on 2015–2019 consumption data will systematically underpredict 2023–2025 demand if Turkish electricity use has been structurally growing due to industrial expansion, rising living standards, and increased AC adoption. Our rolling-origin estimator automatically incorporates the most recent data at each step, which is why our 2024 bias is only 34 MWh while the official is 585 MWh.

Generator strategic behavior. Beyond the TSO, generators themselves have incentives to shade their demand forecasts when submitting market bids. If a generator believes demand will be high (tight supply), bidding a high price is rational. But if all generators believe this and bid high, the clearing price spikes. There is a strategic complementarity: generators want to know what others forecast, not just what actual demand will be. The TSO’s official forecast acts as a coordination device — generators shade their bids relative to it, which may create

feedback dynamics in the market that make the official forecast act as a focal point rather than an accurate estimate.

For the thesis presentation: when an audience member asks “why does the official forecast underpredict?”, the honest answer is: we do not know for certain, but the data is consistent with (a) model staleness in a period of growing demand, and (b) asymmetric institutional loss functions favouring under-procurement. The growing gap in 2024 is most naturally explained by the model not adapting to a structural demand increase compounded by an unusually hot summer.

3 What Is Different About This Data

The raw data is hourly electricity consumption in MWh for each day from 2019 to 2025. The natural object is a **daily load curve**: for each day d , we observe 24 numbers $Y_d(0), Y_d(1), \dots, Y_d(23)$.

The key observation: these 24 numbers are not 24 independent measurements. They are all part of the same daily demand curve — overnight is low, mornings rise steeply, afternoons plateau, evenings surge again. Treating them as 24 separate forecasting problems throws away all of this shape information.

The shape itself is economically meaningful. Two days can have the same *average* consumption but very different *shapes*: one might have a sharp morning peak while another has a flat profile. An air conditioning shock raises the afternoon level but not the overnight level.

Functional Data Analysis (FDA) is the branch of statistics that treats each day’s 24-hour sequence as a single mathematical object — a *curve* — rather than a vector of 24 numbers. This preserves the shape information.

4 Representing the Curve: B-splines

Before doing anything statistical, we need to represent the daily curve mathematically. We use a **B-spline basis**.

Think of it this way. A polynomial approximation says: represent any smooth function $f(h)$ as $\sum_{m=0}^M c_m h^m$. This works but has numerical problems with high-degree polynomials.

B-splines are a better version: piecewise polynomials that are smooth at the joins. We choose $M = 12$ basis functions $B_1(h), \dots, B_{12}(h)$ over the 24-hour domain. Then:

$$Y_d(h) \approx \sum_{m=1}^{12} c_{dm} B_m(h)$$

where the 12 coefficients $c_{d1}, \dots, c_{d,12}$ are chosen to fit the 24 hourly observations for day d . The basis functions $B_m(h)$ are fixed; what changes across days are the coefficients.

What this buys us: instead of working with a jagged sequence of 24 numbers, we work with a smooth curve. We can take its derivative (useful later), integrate it, and apply the tools of functional analysis. The 12 basis coefficients represent the curve accurately because a smooth daily load curve does not need more than 12 degrees of freedom.

5 Functional PCA: Compressing the Curves

5.1 Starting from classical PCA

You likely know classical PCA. Suppose you have n observations of a p -dimensional vector $x_i \in \mathbb{R}^p$. PCA finds the directions of maximum variance: the first principal component ϕ_1 is the direction that explains the most variance in $\{x_i\}$. The score $\xi_{i1} = x_i' \phi_1$ is how far observation i projects onto ϕ_1 .

The decomposition is:

$$x_i = \mu + \xi_{i1}\phi_1 + \xi_{i2}\phi_2 + \cdots + \xi_{ip}\phi_p$$

If the first K components explain most of the variance, we can approximate $x_i \approx \mu + \sum_{k=1}^K \xi_{ik}\phi_k$.

5.2 The functional version

Functional PCA (FPCA) does exactly the same thing but the “observations” are curves $Y_d(\cdot)$ instead of vectors. The “directions” are functions $\phi_k(\cdot)$ called **eigenfunctions** (or functional principal components). The “scores” ξ_{dk} are scalars, exactly as in classical PCA.

The decomposition is:

$$Y_d(h) = \mu(h) + \sum_{k=1}^K \xi_{dk} \phi_k(h) + \varepsilon_d(h)$$

where $\mu(h)$ is the mean curve, $\phi_k(h)$ is the k -th eigenfunction (a 24-hour function), and ξ_{dk} is the score for day d on component k .

What do the eigenfunctions look like in this data?

FPC 1 (explains 93.4% of variance): essentially the “level” component. Its shape is nearly flat across all 24 hours. A high ξ_{d1} means day d has uniformly high consumption. A low ξ_{d1} means uniformly low. Think of it as “how much electricity was used today overall.”

FPC 2 (explains 3.9% of variance): the “shape contrast” component. Its eigenfunction is positive in the morning and evening and negative or flat at midday. A high ξ_{d2} means the morning/evening peaks are pronounced relative to the midday level. A low ξ_{d2} means a flatter profile (more typical of summer when air conditioning smooths the curve).

Together, two components explain 97.3% of the total variation across all 1,096 daily curves. This is the key compression: instead of forecasting 24 numbers, we need to forecast only 2 numbers per day — the two FPCA scores (ξ_{d1}, ξ_{d2}).

5.3 Why $K = 2$?

The choice $K = 2$ follows the standard “95% cumulative variance” rule. Components 3 and 4 explain only 1.1% and 1.0% respectively. Including them would add parameters to the forecasting model without meaningfully improving the functional representation. Parsimony wins.

6 Forecasting the Scores: VAR(1,7)

6.1 The forecasting problem after FPCA

After FPCA, the forecasting problem becomes: given the score history $\{\xi_{d'k} : d' < d\}$ and predetermined features z_d , predict (ξ_{d1}, ξ_{d2}) for tomorrow.

This is a standard multivariate time series problem.

6.2 Vector Autoregression

A **VAR** (Vector AutoRegression) is the multivariate extension of an AR model. If you know AR(1): $y_t = a + \beta y_{t-1} + u_t$. A VAR(1) with $K = 2$ variables is:

$$\begin{pmatrix} \xi_{d1} \\ \xi_{d2} \end{pmatrix} = a + A_1 \begin{pmatrix} \xi_{d-1,1} \\ \xi_{d-1,2} \end{pmatrix} + u_d$$

where A_1 is a 2×2 matrix of coefficients. The VAR allows score 1 to depend on yesterday's score 2, and vice versa.

6.3 Why lags 1 and 7?

Electricity demand has a strong **weekly seasonality**: Monday demand resembles last Monday's, weekends resemble previous weekends. So we include both lag 1 (yesterday) and lag 7 (same weekday last week):

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + u_d$$

This is the FPCA VAR(1,7) — the core dynamic model. It is estimated by OLS equation by equation, which is efficient when the same regressors appear in each equation.

6.4 Adding external features

The VAR alone explains the *dynamics* of scores but ignores known structure: weekends have different demand than weekdays, public holidays cause sharp drops, summer school breaks change the daytime pattern.

We add a vector z_d of predetermined features:

$$\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$$

The features z_d include:

- **Calendar:** weekend indicator, day-of-year sine/cosine, summer school break indicator
- **Holiday:** fixed holidays, Ramadan and Bayram periods, holiday eve, post-holiday, bridge-day indicators
- **Load shape:** yesterday's mean, peak, range, rolling 7-day and 28-day averages
- **Derivative features:** morning ramp, evening ramp, end-of-day drop from the *smoothed derivative* of yesterday's curve

The derivative features deserve special mention. The derivative $DY_{d-1}(h) = \frac{d}{dh} Y_{d-1}(h)$ measures how fast the load curve was changing at each hour yesterday. A steep positive derivative at hour 6–8 means a sharp morning ramp-up. We summarise this by the standard deviation, maximum, minimum, and range of the derivative, plus targeted ramps at specific transition hours. These capture yesterday's *slope behavior* in addition to its level.

Implementation note: all features in z_d are *predetermined* — they are computed from data strictly before day d . There is no look-ahead. This is essential for the evaluation to be honest.

7 Rolling-Origin Evaluation

7.1 The key principle

Evaluating a forecasting model is different from evaluating a regression model. In regression, we split data randomly into train and test sets. In forecasting, the past must precede the future — we cannot train on 2024 data to forecast 2022.

Rolling-origin evaluation (also called walk-forward cross-validation) mimics how the model would be used in practice:

1. Set a test start date: January 1, 2023.
2. For each day d from Jan 1 2023 to Dec 31 2025 (1,096 days):
 - Estimate FPCA on all data strictly before d
 - Estimate the score VAR on all data strictly before d
 - Produce a one-day-ahead forecast for day d
 - Record the forecast error
3. Evaluate across all 1,096 test days.

This design ensures **no information leakage**: the model at each step has access only to the information that would be available in real time. The FPCA eigenfunctions themselves are re-estimated at every step as the training window grows.

7.2 Expanding vs sliding window

We use an **expanding window**: training always uses all available past data. An alternative would be a sliding window (only the last N days). Expanding windows are standard when data is non-stationary but the long-run structure (eigenfunctions, calendar patterns) is stable. We verified that the FPCA eigenfunctions are stable across sub-periods — inner products between 2019–2021 and 2022–2024 eigenfunctions exceed 0.99 — so the expanding window is appropriate.

8 Evaluation Metrics

8.1 The three metrics

For each model m and each (day, hour) pair (d, h) , the forecast error is $e_{dh}^{(m)} = Y_d(h) - \hat{Y}_d^{(m)}(h)$.

$$\begin{aligned} \text{MAE}_m &= \frac{1}{N} \sum_{d,h} |e_{dh}^{(m)}| \\ \text{RMSE}_m &= \sqrt{\frac{1}{N} \sum_{d,h} (e_{dh}^{(m)})^2} \\ \text{MAPE}_m &= \frac{1}{N} \sum_{d,h} \frac{|e_{dh}^{(m)}|}{Y_d(h)} \times 100 \end{aligned}$$

MAE is the average absolute error in MWh. Directly interpretable in the units of the problem.

RMSE penalises large errors more (because of the square). If grid operators care especially about avoiding large misses on peak days, RMSE is the more relevant metric. Note that OLS minimises RMSE in-sample, so RMSE is also the “coherent” metric for OLS.

MAPE is percentage error. It is scale-free, which matters when comparing across years: if Turkish electricity demand grows 5% between 2023 and 2025, a fixed MAE threshold becomes harder to beat in later years. MAPE adjusts for this. In the electricity load forecasting literature, MAPE is the most commonly reported metric.

8.2 Relative gain vs the official forecast

We also report the **gain** relative to the official forecast:

$$\text{MAE gain}_m = 100 \cdot \frac{\text{MAE}_{\text{off}} - \text{MAE}_m}{\text{MAE}_{\text{off}}}$$

Positive gain = model m beats the official forecast. Negative gain = model m is worse.

8.3 The results

Model	MAE	RMSE	MAPE	MAE gain	RMSE gain	MAPE gain
Seasonal naive Y_{d-7}	1912	3130	5.13%	−57.6%	−79.8%	−62.9%
Official forecast	1214	1740	3.15%	0%	0%	0%
FPCA VAR(1,7) scores only	1443	2018	3.76%	−18.9%	−16.0%	−19.4%
+ ramp/calendar	1238	1716	3.26%	−2.1%	+1.4%	−3.7%
+ holiday/load regime	1001	1354	2.63%	+17.5%	+22.2%	+16.5%
+ derivative features	980	1324	2.56%	+19.2%	+23.9%	+18.7%
+ residual step	653	983	1.70%	+46.2%	+43.5%	+46.0%

Reading the table: the model-building path shows each addition. FPCA VAR(1,7) alone is *worse* than the official forecast — the pure dynamics without any calendar information are not enough. The ramp/calendar model barely helps. The big jump happens when we add holiday and load-shape information: MAE drops from 1238 to 1001. Derivative features add a further push to 980.

9 Why Comparing MAE Numbers Is Not Enough: The DM Test

9.1 The problem with just looking at numbers

Suppose model A has MAE = 980 and model B has MAE = 990. Model A looks better. But is this difference real, or could it arise from random variation in the 1,096-day test period?

This is a standard statistical question: we need a hypothesis test.

9.2 The Diebold-Mariano test

The **Diebold-Mariano (DM) test** (Diebold & Mariano, 1995) answers exactly this question. Here is the logic:

Define the **loss differential** for each observation t :

$$d_t = |e_t^{(\text{challenger})}| - |e_t^{(\text{official})}|$$

If $d_t < 0$, the challenger won on observation t . If $d_t > 0$, the official won. The null hypothesis is $\mathbb{E}[d_t] = 0$ (equal expected loss). The alternative is $\mathbb{E}[d_t] < 0$ (challenger has lower expected loss).

The DM statistic is essentially a t -test on $\bar{d}$, the sample mean of the loss differentials, with a heteroskedasticity-and-autocorrelation consistent (HAC) variance estimate. We use the Harvey, Leybourne & Newbold (1997) small-sample correction.

Connection to what you know: the DM test is a t -test on the time-series mean of d_t , using a long-run variance estimator (like a Newey-West standard error in your metrics courses). It accounts for serial correlation in the errors, which matters because consecutive forecast errors tend to be correlated.

9.3 The results

Challenger	$\bar{d}$ (MWh)	DM stat	p -value	Beats official?
FPCA VAR(1,7) scores only	+229	+22.9	≈ 1	No — significantly worse
+ ramp/calendar	+25	+2.6	0.996	No
+ holiday/load regime	−213	−24.9	10^{-135}	Yes, $p < 0.001$
+ derivative features	−233	−27.4	10^{-163}	Yes, $p < 0.001$
+ residual step	−561	−71.2	≈ 0	Yes, dominant

The p -value of 10^{-163} for the main model means the probability of observing this large an improvement by chance, if both models were truly equal, is essentially zero.

We also run the DM test **hour by hour** (24 separate tests). The main model beats the official forecast at 19 of 24 hours, and the difference is statistically significant (5% level) at 18 of 24.

The 5 hours where the official forecast wins are hours 6, 7, 8 (morning ramp-up) and 18, 19 (evening surge). These are the transition hours where the load curve changes fastest. At hour 8, the MAPE loss is −24% — the official is 24% more accurate. This is likely because the official forecast incorporates temperature and weather information that determines the *timing* of the ramp; our model uses only historical load patterns.

10 ACF Diagnostics: Are the Errors Random?

10.1 What the ACF measures

After fitting any model, we ask: are the residuals white noise? If errors follow a systematic pattern, the model has missed something that could in principle be captured.

The **autocorrelation function (ACF)** measures:

$$\rho_k = \text{Corr}(e_t, e_{t-k})$$

the correlation between today’s error and the error k days ago. If the model is correctly specified, all ρ_k should be near zero.

The **Ljung-Box test** is a formal test of: $H_0 : \rho_1 = \rho_2 = \dots = \rho_{14} = 0$ (no autocorrelation up to lag 14). Rejection means remaining serial structure.

10.2 What we find

VAR score residuals (in-sample): fitting the pure VAR(1,7) without external features on the 2019–2022 training data, the Ljung-Box statistic is 504 for Score 1 and 553 for Score 2 (both $p = 0$). This means the scores have remaining autocorrelation beyond lags 1 and 7 — which is expected, because holidays and seasonality create systematic patterns in the scores that the VAR without features cannot capture. This is the *motivation* for adding external features.

Curve forecast errors (out-of-sample): for the full main model with all features, the Ljung-Box statistic at each of the 24 hours ranges from 197 to 7,227, all with $p = 0$. This means even the best model leaves strongly autocorrelated errors.

Why are the out-of-sample errors so autocorrelated? Look at the distribution across hours. Hour 6 has LB = 7,227. This is the morning ramp start. The model systematically over-forecasts or under-forecasts the ramp timing for consecutive days — during holiday periods the ramp shifts to a different pattern and persists for days; during summer the ramp is shallower and this also persists. These regime patterns create serial dependence in

errors that cannot be removed by lagged features alone.

10.3 Why this matters: the residual step

The key insight is: *if errors are predictable, we can correct for them*. The **residual second step** does exactly this. For target day d , after producing the base forecast $\hat{Y}_d^{base}(h)$:

1. Compute past fitted residuals: $\hat{e}_s(h) = Y_s(h) - \hat{Y}_s^{base}(h)$ for all $s < d$ (strictly past data only).
2. Estimate a correction curve $\hat{r}_d(h)$ from recent past residuals.
3. Apply the correction: $\hat{Y}_d^{res}(h) = \hat{Y}_d^{base}(h) + \lambda_d \hat{r}_d(h)$ where λ_d is chosen by rolling-origin cross-validation.

This is rolling-origin throughout: no information leakage. The massive error autocorrelation — especially at ramp hours — is exactly what the residual step leverages. It wins at all 24 of 24 hours including the 5 hours where the base model loses to the official forecast.

11 Eigenfunction Stability and Dynamic FPCA

A natural question about FPCA is: do the eigenfunctions ϕ_1, ϕ_2 remain stable over time, or do they shift as the Turkish grid evolves? If they shift, we might need **dynamic FPCA** — a more complex method that allows time-varying eigenfunctions.

We tested this directly:

1. Fit FPCA separately on three sub-periods: 2019–2021, 2020–2022, 2022–2024.
2. Compute the **inner product** between eigenfunctions from different periods. An inner product of 1 means identical shape; 0 means orthogonal (completely rotated).

Results: all pairwise inner products exceed 0.993. The variance proportion explained by FPC1 stays between 92.4% and 93.2% across all sub-periods. The eigenfunctions are stable.

Conclusion: dynamic FPCA is not needed for this dataset. Turkish electricity load shapes — the overnight trough, morning ramp, midday plateau, evening peak — have not structurally changed over 2019–2024. Static FPCA is appropriate, and we can say so with empirical evidence rather than just assumption.

12 The Random Forest Benchmark

12.1 Why a machine learning comparison?

The score regression in the main model is linear OLS: $\xi_d = a + A_1 \xi_{d-1} + A_7 \xi_{d-7} + B z_d + u_d$.

A natural question: does the relationship between the features z_d and the scores ξ_d have non-linear structure that OLS misses?

Random forest (RF) answers this. RF on the same feature vector z_d plus lagged scores can capture any smooth nonlinear function of the inputs without specifying the functional form.

12.2 Why keep the FPCA structure

We apply RF to the *score regression* only — we still use FPCA to represent the curves and reconstruct the forecast curve from predicted scores. This keeps the functional data analysis framework intact. The RF is a drop-in replacement for OLS, not an alternative to FPCA.

12.3 What the comparison tells us

- If **FPCA-OLS** $\approx$ **FPCA-RF**: the linear score equation is adequate. We get the same accuracy with a structured, interpretable model. This is the preferred outcome for the thesis’s FDA identity.
- If **FPCA-RF** $\gg$ **FPCA-OLS**: there is meaningful nonlinearity in the feature-score relationship. This motivates a future extension but does not invalidate the current results.

12.4 Results

FPCA-RF vs FPCA-OLS (full 2023–2025 evaluation). Using the same feature vector z_d plus lagged scores as inputs to a random forest (one RF model per FPCA score, same rolling-origin design, FPCA basis fixed on the expanding training window):

Model	MAE	MAPE	MAE gain	Conclusion
Official forecast	1214	3.15%	0%	benchmark
FPCA-RF (fixed basis)	1217	3.14%	−0.3%	ties official
FPCA-OLS (main model)	980	2.56%	+19.2%	best

The outcome is the first case: **FPCA-OLS substantially outperforms FPCA-RF**. RF achieves essentially the same accuracy as the official forecast (MAE 1217 vs 1214) while OLS beats the official by 19%. RF finds no meaningful nonlinearity to exploit: the linear score equation is adequate, and the VAR structure plus linear feature mapping is actually beneficial as a regulariser.

Conclusion: The FPCA scores are approximately linearly predictable from the features. OLS’s linear structure reduces variance without sacrificing fit; RF’s additional flexibility adds variance without a compensating reduction in bias. This validates the linear score regression and the FDA/VAR model architecture.

Direct RF as an additional exercise (2025 subsample only). A “direct RF” model that skips the FPCA structure entirely and predicts each of the 24 hourly values directly (using the same features plus lagged hourly values) achieves MAE = 718 on the 2025 subsample, a 31.8% gain over the official (MAE 1052) in that year, compared to FPCA-OLS’s 13.5% gain (MAE 910) in the same period. A direct ML approach can therefore be competitive, but the comparison is on a single year and does not use the functional structure. The FPCA model’s 19.2% gain over the full three-year period — with interpretable eigenfunctions, a testable stability condition, and a clear decomposition of the forecast — remains the primary result.

13 The Coherent Narrative

Here is the full story in one flow:

1. **The object:** daily electricity consumption is a curve, not a vector. Treating it as a curve preserves intraday shape.
2. **The compression:** FPCA reduces 24 correlated observations to 2 uncorrelated scores that explain 97.3% of curve variation. The eigenfunctions are empirically stable over 6 years.
3. **The dynamics:** scores follow a VAR(1,7) with calendar, holiday, load-shape, and derivative features. The features are essential: without them, the model is worse than the official forecast.

4. **The gain:** the main model ($K=2$, all features) beats the official forecast by 19.2% MAE and 18.7% MAPE, with a Diebold-Mariano p-value of 10^{-163} . The improvement is real.
5. **Where it loses:** transition hours (6–8, 18–19) where the official forecast’s weather information gives it an edge. The DM test shows this explicitly.
6. **The residual step:** ACF diagnostics show that errors are strongly autocorrelated, especially at transition hours. The residual step exploits this predictable pattern and achieves 46% MAE gain — winning at all 24 of 24 hours.
7. **The benchmark:** FPCA-RF (fixed basis) achieves $\text{MAE} = 1217$, essentially tying the official forecast, while FPCA-OLS achieves $\text{MAE} = 980$. The linear score equation is adequate; RF finds no nonlinearity to exploit.
8. **Statistical validity:** rolling-origin evaluation ensures no leakage. DM tests confirm the gains are statistically significant. ACF diagnostics characterise the remaining structure.

14 Concepts to Know Cold for the Presentation

Term	One-sentence explanation
B-spline	Piecewise polynomial basis for approximating smooth functions
FPCA	PCA applied to curves; gives eigenfunctions and daily scores
Score	The projection of a curve onto an eigenfunction; a scalar
Eigenfunction	The “principal component” in function space; a 24-hour curve
VAR(1,7)	Multivariate AR with lags 1 and 7; estimated by OLS equation by equation
Rolling-origin	Forecasting analogue of cross-validation; train only on strictly past data
MAE	Mean absolute error in MWh; primary accuracy metric
MAPE	Mean absolute percentage error; scale-free version of MAE
RMSE	Root mean squared error; penalises large errors more
DM test	Hypothesis test for equal forecast accuracy; t -test on loss differential
ACF	Autocorrelation function; measures serial dependence in residuals
Ljung-Box	Portmanteau test: are ACFs jointly zero up to lag q ?
Residual step	Second-stage correction using past errors; exploits error autocorrelation
Expanding window	Training set grows as new data arrives; appropriate when structure is stable
Inner product	Cosine similarity between two eigenfunctions; 1 = identical, 0 = orthogonal